





# Day 3: Covariate adjustment with quantitative outcome, ANCOVA, robustness, missing data, MMRM

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# Outline/Intended Learning Outcomes (ILOs)

## Covariate adjustment

- ILO: repeat the main recommendations
- ILO: explain how this can lead to power gains
- ILO: restate key robustness properties
- ILO: perform conventional analyses for quantitative outcomes
- ILO: recall examples

## Missing data, with a focus on baseline follow-up studies

- ILO: list a few principles and type of missing data
- ILO: fit a Mixed Model for Repeated Measurements (MMRM) and interpret the results
- ILO: restate the rationale for using an MMRM analysis



# Covariate adjustment: quantitative outcomes (1/2)

- Instead of a multiple logistic model to analyse binary outcomes, one can use a multiple linear model to analyse quantitative outcomes (aka ANCOVA[1]).
- *"The resulting estimated regression coefficient for the treatment indicator is the estimate of the average treatment effect."* [2] (when the is no baseline $\times$ treatment interaction terms; otherwise centering or other tricks might be needed.)
- Similar robustness and power gain properties exist, as those already seen with binary outcome analyzed via logistic regression:

  *"Even when the linear regression model is misspecified and does not accurately capture the relationships between the outcome, covariates, and treatment, covariate adjustment through a linear model is a valid method for estimating and performing inference for the average treatment effect."*

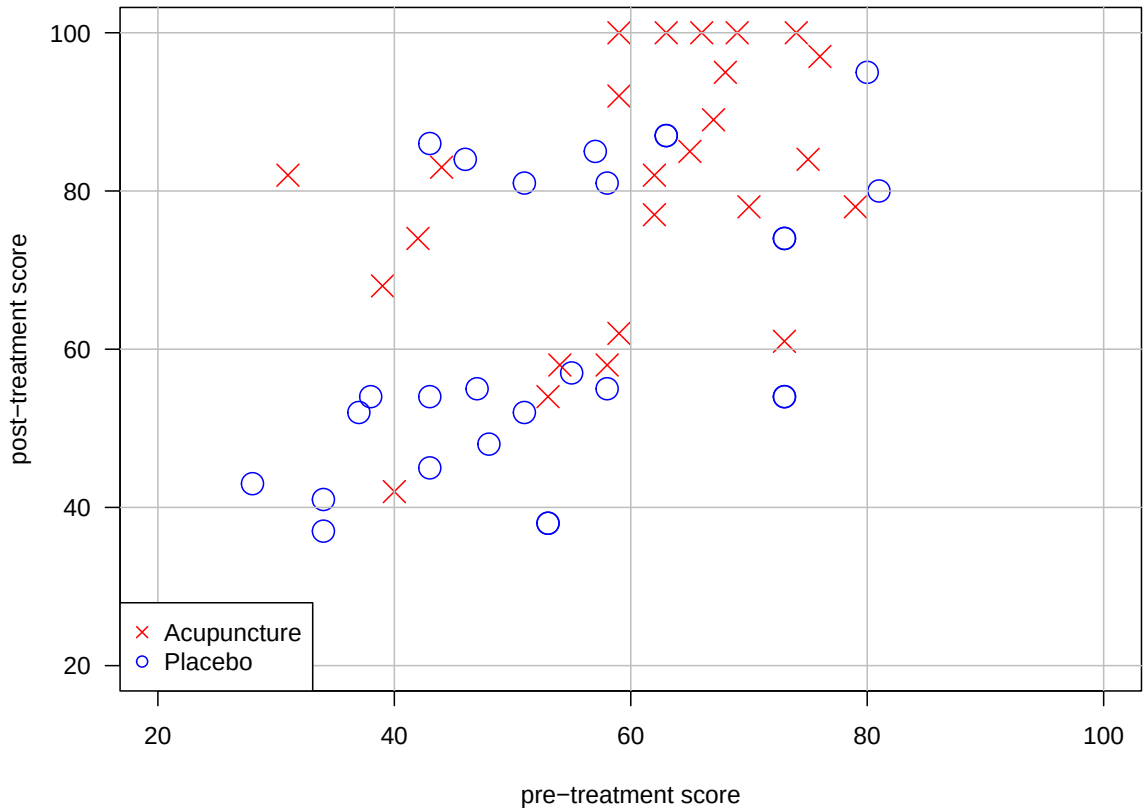
- Interestingly, the power gains obtained by covariates adjustment (or equivalently reduction in sample size needed) are often more substantial with quantitative outcomes than with binary outcomes!

[1] ANCOVA stands for ANalysis of COVAriance

[2] FDA scientific guidelines: "Adjusting for Covariates in Randomized Clinical Trials for Drugs and Biological Products Guidance for Industry", 2023, accessed 29/10/2025, https://www.fda.gov/media/148910/download

# Intuition & Example: baseline-follow up studies

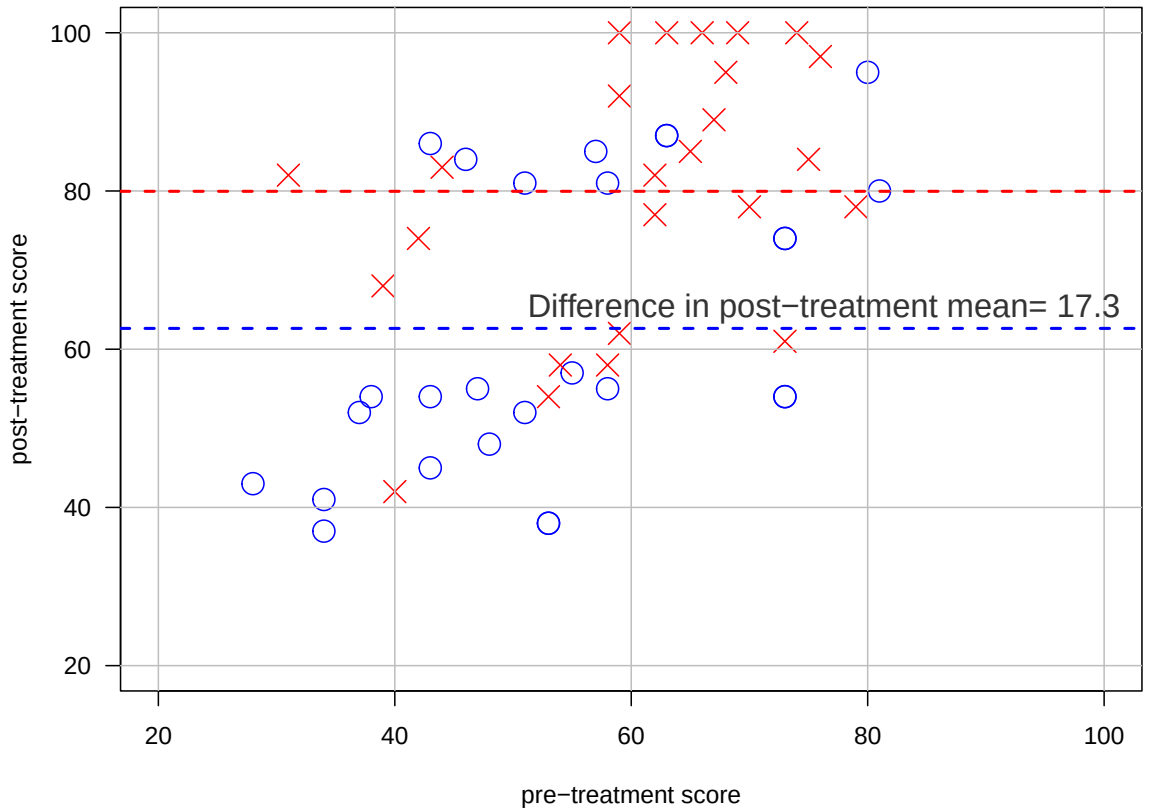
Clear association between baseline and follow-up scores, within each arm: the larger the pre-treatment score the larger post-treatment score.




Data: reconstructed patient level data nearly perfectly matching those graphically presented in Vickers & Altman. BMJ. 2001 Nov 10;323(7321):1123-4.

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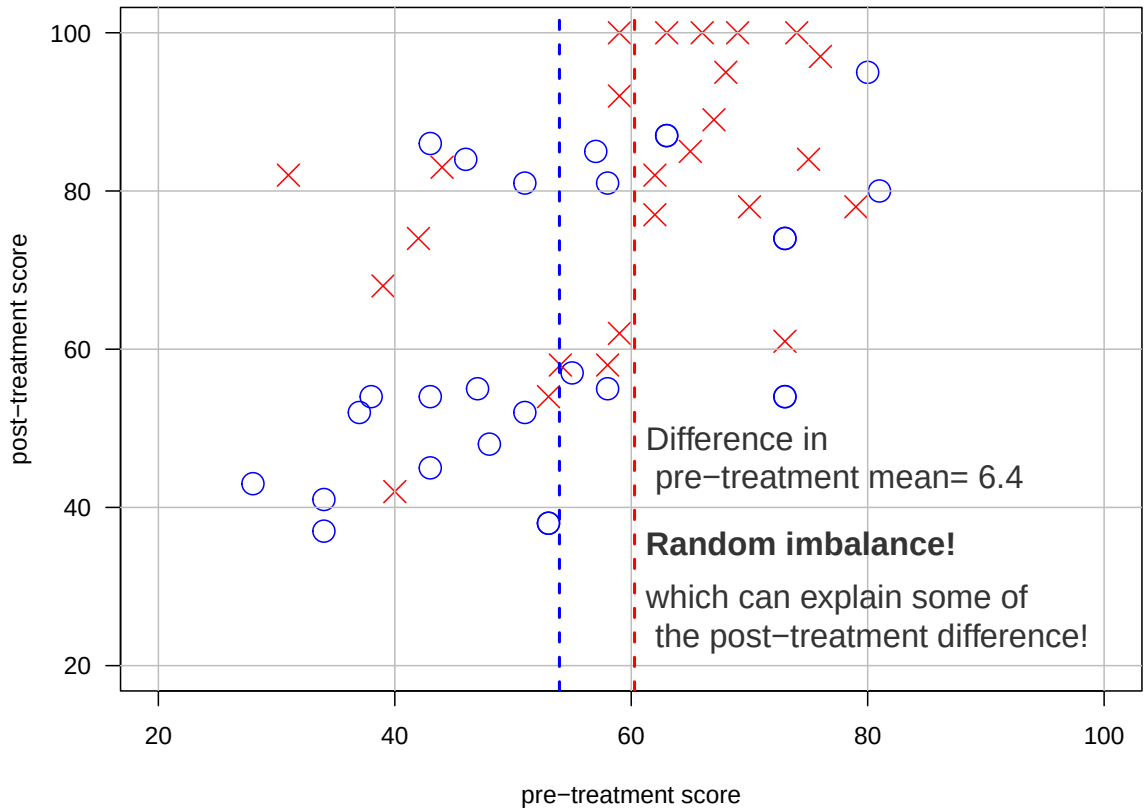
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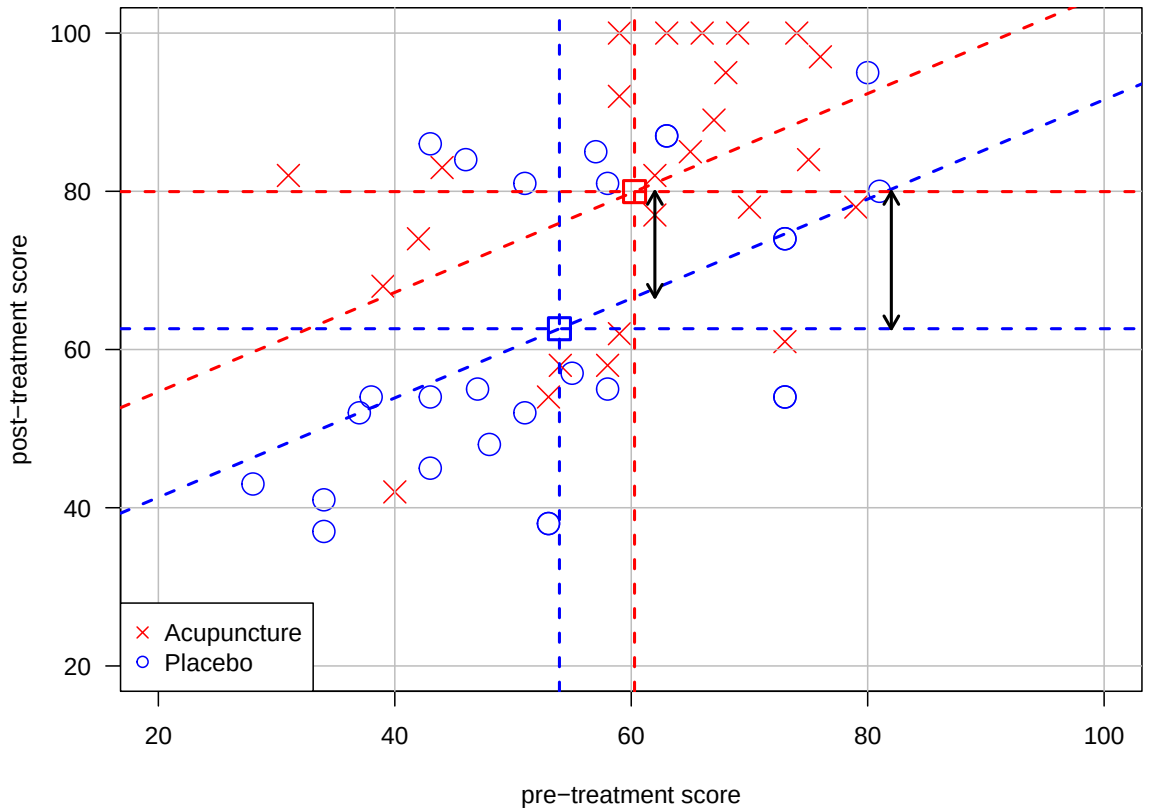
# Intuition & Example: baseline-follow up studies

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# Intuition & Example: baseline-follow up studies (2/2)




The multiple linear model, adjusted for pre-treatment score, will always correct/adjust for random imbalance in baseline scores. Implicitly, it estimates the mean difference in post-treatment score as if the mean pre-treatment score were the same in the two arms. This reduces the variability of the estimated difference, since random imbalance makes the unadjusted estimate randomly too small or randomly too large. Hence we can compute smaller standard errors for the adjusted differences. This leads to narrower confidence intervals and smaller p-values, in average (but in average only!).

# Common questions

- How much power can we gain when using an adjusted analysis (i.e., ANCOVA) instead of a simple t-test?

  Potentially a lot! The larger the correlation between the pre and post scores and (i.e., the steeper the slope) and the larger the power we gain. In terms of sample size needed, one might actually use $\sigma' = \sigma\sqrt{1-\rho^2}$ instead of $\sigma$ in sample size calculation formulas, where $\sigma$ is the SD of the post-treatment score and $\rho$ the correlation.

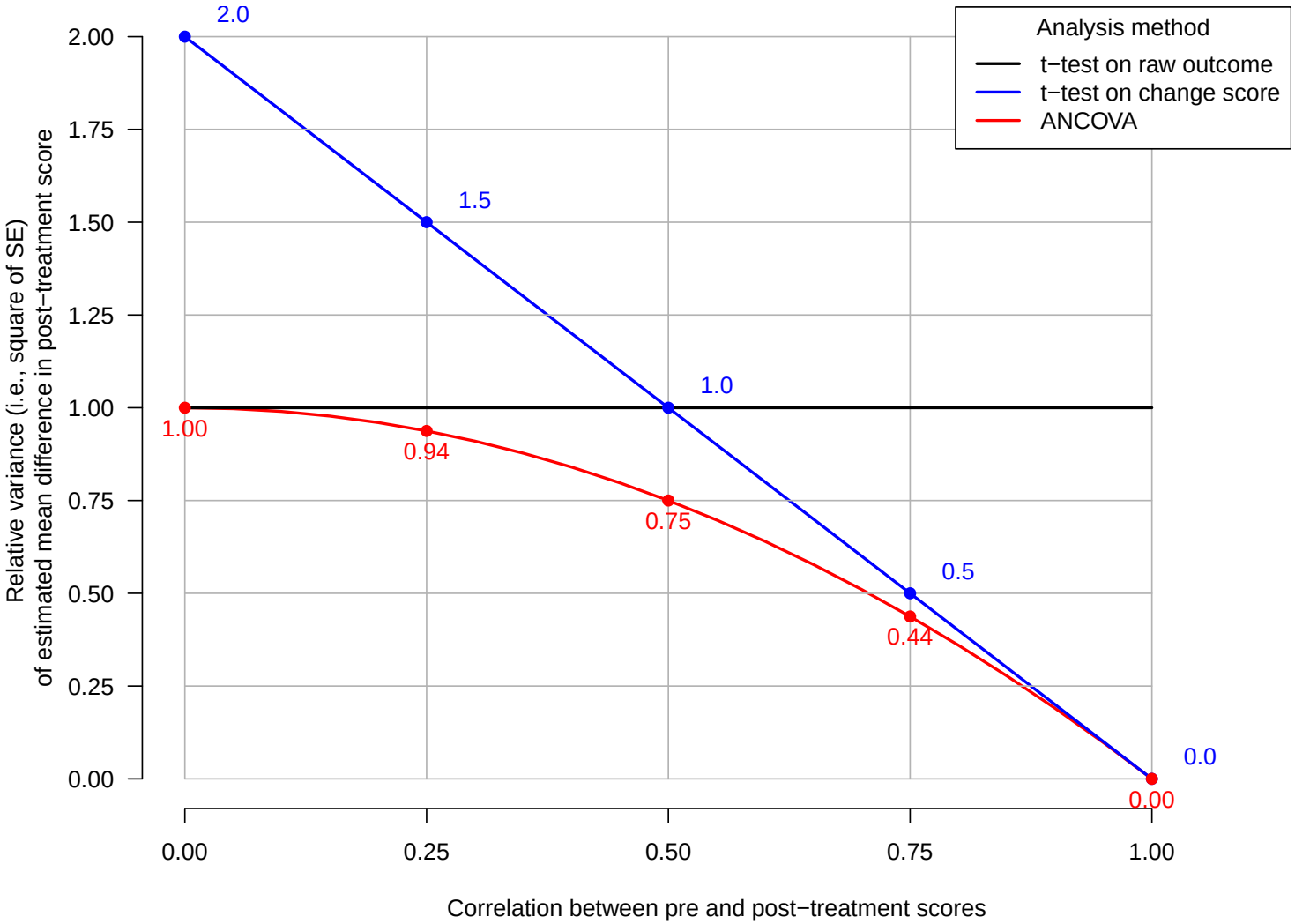
- Isn't it sufficient to adjust for baseline by comparing the change scores (i.e., difference between pre and post treatment score)? Isn't it even better than ANCOVA?

  No! It is not as good as using an ANCOVA, and sometimes worse than using a t-test to compare post-treatment scores (in case of weak correlation, when $\rho < 0.5$).

Details: see e.g., Section 7.2.3, in *Statistical Issues in Drug Development* ($3^{rd}$ Edition), by Stephen Senn.



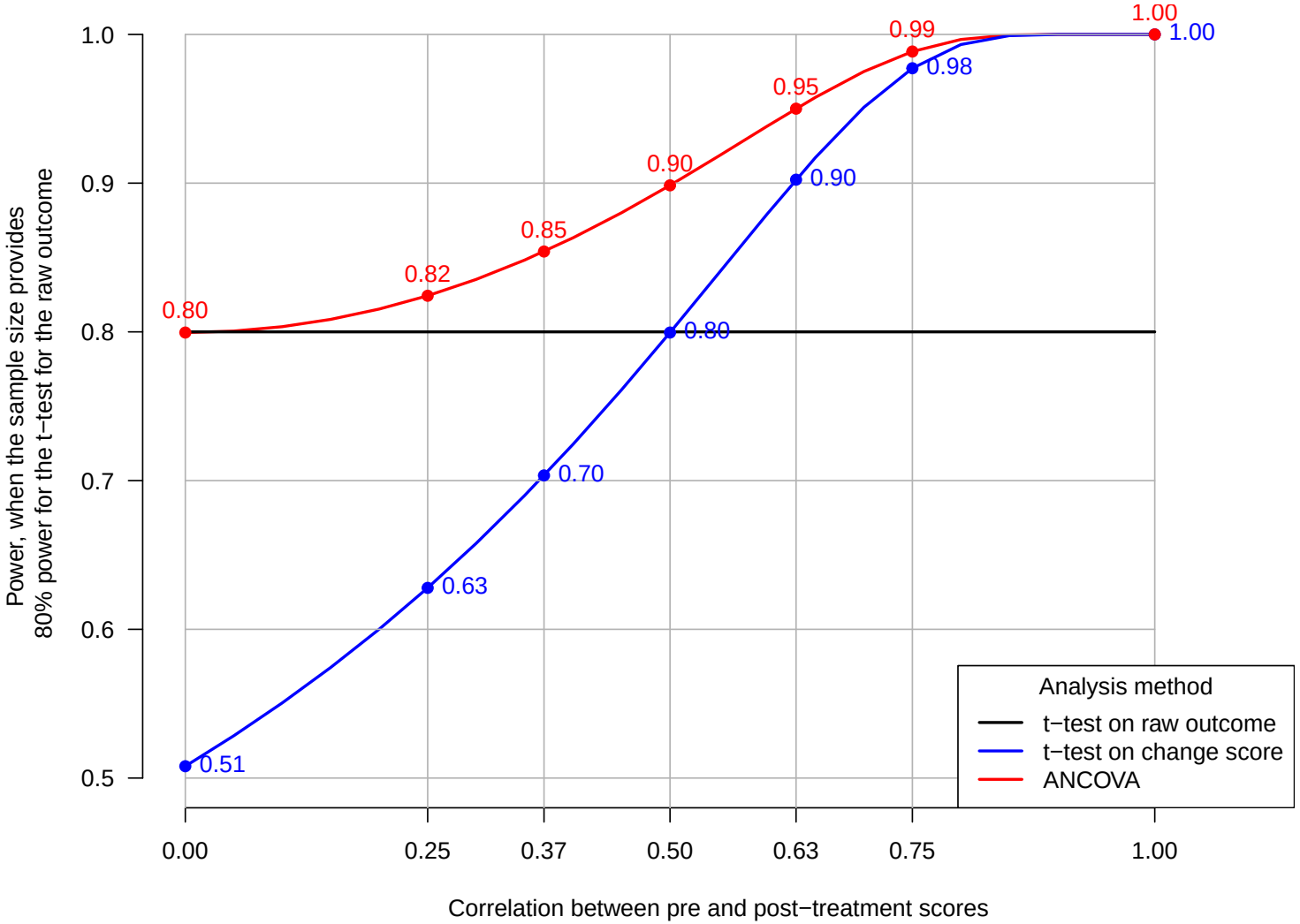
# More detailed answers (1/2)




Note: "large" sample, but usually very accurate approximation (assumming same variability at baseline and follow-up).

Remark: relative variance equals relative sample size needed to have the same power.

# More detailed answers (2/2)




Note: "large" sample, but usually very accurate approximation (assumming same variability at baseline and follow-up).

# Remarks

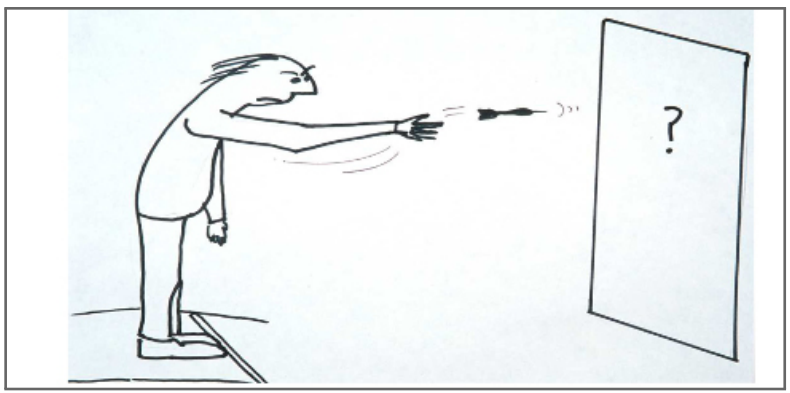
- The same "intuition" and "mathematics" work when adjusting for more than one baseline covariate. Usually, we use $5 \pm 2$, that we expect are (altogether) the most predictive of the outcome. But it really depends on the sample size and the background knowledge. Each additional covariate need to add "non-negligible" prognostic information on top of the others, to be useful. There is no simple rule!
- Power gains are often substantially larger for quantitative outcomes (when using ANCOVA) than whith binary outcomes (when using logistic regression, with or without standardization).
- When adjusting for more than one covariate, formulas such as $\sigma' = \sigma\sqrt{1-\rho^2}$ are becoming more difficult to use. Hence, conservative choices are usually made (e.g., sample size and power are computed as if only one covariate only will be adjusted for, although more covariates will be adjusted for).
- If the prognostic ability of baseline covariates is unclear / unknown, it may be wise to be conservative; this is a common and well accepted choice:

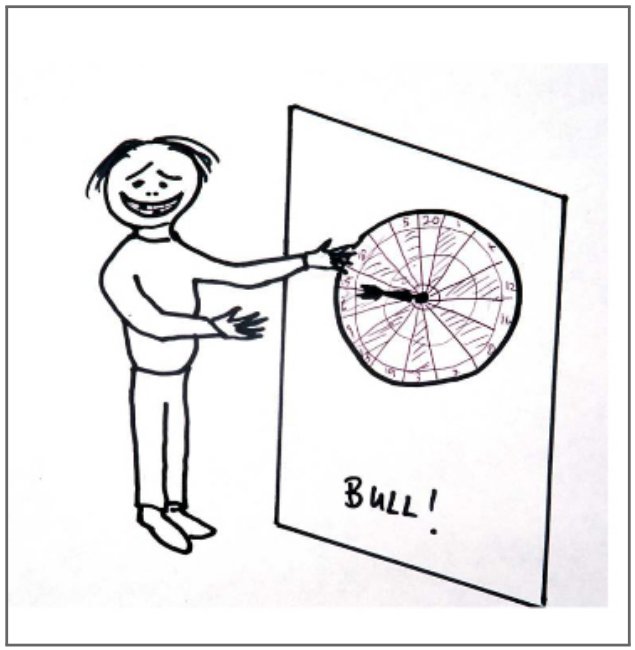
> *"In a trial that uses covariate adjustment, the sample size and power calculations can be based on adjusted or unadjusted methods. The latter will often lead to a more conservati sample size."*[3]



[3] FDA scientific guidelines: "Adjusting for Covariates in Randomized Clinical Trials for Drugs and Biological Products Guidance for Industry", 2023, accessed 29/10/2025, https://www.fda.gov/media/148910/download

# Digression: prespecification matters!




Concluding significance without prespecification is like drawing a dart-board around where the dart lands.

Of course, pre-specification of whether we plan to use an ANCOVA, a t-test or a t-test on change score is essential. We cannot perform the three analyses and report the best looking results!

# Digression: three results with the acupuncture trial

| Analysis Method | Est. ($\hat{\delta}$) | 95%-CI | SE | p-value ($\mathcal{H}_0 : \delta = 0$) |
|---|---|---|---|---|
| ANOVA | 13.3 | [4.4;22.3] | 4.4 | 0.42% |
| t-test (on post-treatment score) | 17.3 | [7.5;27.2] | 4.9 | 0.09% |
| t-test on change score | 11.0 | [1.9;20.0]] | 4.5 | 1.86% |

Note: here $\hat{\rho} = 0.53$, baseline imbalance (difference in pre-treatment mean)$=$ 6.4.

Remark: The recommended approach (ANCOVA) had the smallest SE (as expected) but not the smallest p-value (unlike what is expected "on average" or "the most often") because of random imbalance at baseline.

Remark: comparing change scores or post-treatment scores are two approaches that both target the same treatment effect, as in average there is no difference pre-treatment thanks to randomization. In other words, the mean difference post-treatment (at the population level) that we aim to estimate is equal to the mean difference in change score, as a consequence of randomization.



# Exercise 3.1

1. Practice with R! Run the code of the script "`acunpunctureRDemo.R`" and recognize the main results of the previous slides.

2. Many researchers wonder whether they should define their primary outcome as the score at end of follow-up (i.e., post-treatment score) or as the change score at end of follow-up (i.e., post-treatment score minus pre-treatment score). Do you think it matters, if the primary analysis is made using an ANCOVA to adjust for the baseline score? Think...

3. Perform the ANCOVA using the change from baseline as the outcome. What do you observe?

Remark: what you have just observed on this specific dataset always holds. It is an old (but important!) result[4], which is relatively well-known.





[4] See e.g., Section 7.2.4, in *Statistical Issues in Drug Development* ($3^{rd}$ Edition), by Stephen Senn, in which there is a reference to the famous paper of Laird (1983, *The American Statistician* 37: 329–330).

# A recent example (1/4) (Danish investigator-initiated RCT, completed in 2025)

## Early discontinuation of empirical antibiotics versus extended treatment in children with cancer and high-risk febrile neutropenia in Denmark: an open-label, randomised controlled trial

*Nadja H Vissing, Torjus Skajaa, Dorthe Grosen, Birgitte Klug Albertsen, Bodil Als-Nielsen, Henrik Hasle, Milos M Koch, Annett H Rasmussen, Jesper S Brok, Lisa Hjalgrim, Paul Blanche, Kjeld Schmiegelow, Ulrikka Nygaard*

### Summary

**Background** Early discontinuation of empirical antibiotics in adults with high-risk febrile neutropenia based on clinical stability and defervescence, irrespective of neutrophil count, has been shown to reduce antibiotic exposure without compromising safety. We aimed to evaluate this approach in children with cancer, a population for which randomised trials in high-income settings are lacking.

**Methods** We conducted an investigator-initiated, multicentre, open-label, randomised, superiority trial in three paediatric haematology–oncology centres in Denmark. Patients aged 0–17 years receiving intensive chemotherapy and presenting with fever during high-risk neutropenia (absolute neutrophil count $<0{\cdot}5\times10^9$ cells per L anticipated to last ≥7 additional days) were eligible. Patients with microbiologically documented infection were excluded. Fever episodes were randomly assigned (1:1) to either early discontinuation of intravenous antibiotics after 48 h of defervescence and clinical stability (short treatment group), or continuation until neutrophil recovery with a maximum of 10 days per episode (long treatment group). Randomisation was done using a computer-generated sequence embedded in a dedicated electronic data capture system, stratified by site, via block randomisation with random block size (of four, six, or eight). The randomisation sequence was concealed until allocation, and investigators were unaware of group assignment until completion of randomisation. The primary endpoint was the number of days on antibiotics within 28 days after the first treatment initiation. Main analyses were done in the intention-to-treat population. Safety was assessed by infection-related serious adverse events, including bacteraemia, focal infection, sepsis, and death. The trial was registered with ClinicalTrials.gov, NCT04637464, and is completed.

*Lancet Child Adolesc Health* 2026

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# A recent example (2/4) (Danish investigator-initiated RCT, completed in 2025)

For the primary outcome, we estimated the between-group difference in mean number of days on antibiotics using a multiple linear regression model, fitted with generalised estimating equations to account for potential within-patient correlation between episodes. Covariates included in the model were age at random assignment, cancer type (acute myeloid leukaemia or relapsed acute lymphoblastic leukaemia, acute lymphoblastic leukaemia during induction, acute lymphoblastic leukaemia after induction, or other), ANC at presentation ($\leq 0 \cdot 1 \times 10^9$ cells per L, $>0 \cdot 1$ to $<0 \cdot 5 \times 10^9$ cells per L, or $\geq 0 \cdot 5 \times 10^9$ cells per L), platelet count at presentation, and study site. To account for potential non-linearity, the effects of age and platelet count were modelled using linear splines.

# A recent example (3/4) (Danish investigator-initiated RCT, completed in 2025)

*"There was a random imbalance in cancer type and sex between the two groups. The number of children with acute myeloid leukaemia and relapsed acute lympho blastic leukaemia was lower in the short treatment group than in the long treatment group, and the proportion of children with other cancer types was higher in the short treatment group than the long treatment group (table 1)."*

- And cancer type is an important prognostic covariate!

| | Short treatment group (n=45) | Long treatment group (n=43) |
|---|---|---|
| Sex | | |
| Female | 24 (53%) | 13 (30%) |
| Male | 21 (47%) | 30 (70%) |
| Site | | |
| Copenhagen | 29 (64%) | 31 (72%) |
| Odense | 6 (13%) | 3 (7%) |
| Aarhus | 10 (22%) | 9 (21%) |
| Age, years | 4·8 (3·2–8·4) | 5·8 (3·5–9·7) |
| Year of inclusion | | |
| 2020 | 1 (2%) | 1 (2%) |
| 2021 | 10 (22%) | 11 (26%) |
| 2022 | 13 (29%) | 13 (30%) |
| 2023 | 15 (33%) | 13 (30%) |
| 2024 | 5 (11%) | 5 (12%) |
| 2025 | 1 (2%) | 0 |
| Cancer type* | | |
| Acute myeloid leukaemia | 6 (13%) | 8 (19%) |
| Acute lymphoblastic leukaemia | 33 (73%) | 33 (77%) |
| Relapse | 2 (4%) | 7 (16%) |
| De novo, induction | 5 (11%) | 4 (12%) |
| De novo, after induction | 26 (58%) | 22 (47%) |
| Other cancers | 6 (13%) | 2 (5%) |
| Days from cancer diagnosis to randomisation | 85 (54–181) | 71 (47–128) |

**Source:** Vissing et al. The Lancet Child & Adolescent Health, 2026; 10, 418-428

# A recent example (4/4) (Danish investigator-initiated RCT, completed in 2025)

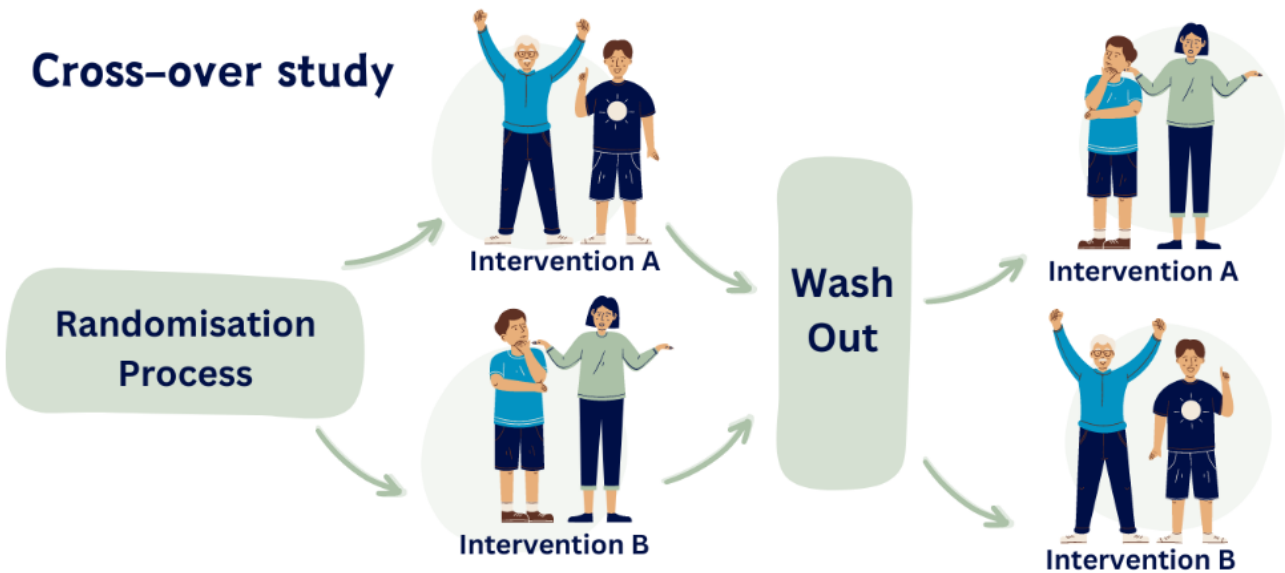
> In the primary analysis, the time on antibiotic treatment within the 28-day follow-up was a mean of 8·1 days (SD 5·9) in the short treatment group versus 13·5 days (6·2) in the long treatment group, leading to an estimated difference of –4·0 days (95% CI –6·0 to –1·9; p=0·0002, adjusted for baseline covariates). The course of febrile episodes is shown in figure 2. The proportion of febrile episodes for which antibiotic treatment was administered on each day during follow-up is shown in figure 3. The unadjusted difference was –5·5 days (–8·0 to –2·9; p<0·0001). The per-protocol analysis showed highly similar results to the intention-to-treat analysis (appendix p 4).

- Adjusted analysis lead to
  1. reduced treatment effect estimate (corrected for random imbalance)
  2. narrower 95%-CI: width of $6.0 - 1.9 = 4.1$ vs $8.0 - 2.9 = 5.1$ (more precision/power, as random imbalance is systematically corrected for)

  as compared to the unadjusted analysis provided for completeness.

# Digression: why are crossover designs useful? (1/2)

*In a crossover study, each participant receives more than one treatment at different times. Participants are first given one treatment and then after a washout period they change to the other treatment. This study design sees every person acting as their own comparison, which in turn helps the researchers better understand the effects of each treatment.*




Source: https://unimelb.libguides.com/whichstudytype/Cross-over, accessed Nov 4, 2025.

# Digression: why are crossover designs useful? (2/2)

## A short answer:

*"The within-patient aspect of the design eliminates a staggering number of potential confounders:*[5] *perhaps upward of 20,000 genes, any epigenetic factors, and the whole environmental history of the subject up until the start of the trial. The analysis could not possibly account for these factors individually but does not have to. Their joint effect as confounders is eliminated by subtracting outcomes under placebo from those under ISF240* [i.e., experimental treatment] *to form paired differences, which are the basis for analysis."* [6]

[5] Here 'confounders' refers to prognostic baseline variables that will inevitably be randomly imbalance, to some extent, despite randomization.

[6] Senn. Journal of Clinical Epidemiology 148 (2022): 184-188.

# Stratified randomization and adjustment

- *"Randomization is often stratified by baseline covariates. A covariate adjustment model should generally include strata variables and can also include covariates not used for stratifying randomization."* [7]

- Not including strata variables usually lead to confidence intervals that are too wide, type-I error rates that are too low and a reduction in power.[8]

- Although it certainly reduces random imbalance, stratified randomization does not provide more power than simple randomization, when adjusting on strata variables in any case (in "large" trials; asymptotic results). In other words, to maximize the power, covariate adjustment matters, but stratified randomization does not.[9]

[7] FDA scientific guidelines: "Adjusting for Covariates in Randomized Clinical Trials for Drugs and Biological Products Guidance for Industry", 2023, accessed 29/10/2025, `https://www.fda.gov/media/148910/download`

[8] See e.g., Kahan and Morris (2012). Statistics in Medicine, 31(4):328–340.

[9] See e.g., Wang et al. (2021). Journal of the American Statistical Association, 118(542), 1152–1163.

# Further remarks (1/2)

- Do not adjust on post-randomization variables. Only use baseline variables (i.e., those known at time of randomization). Otherwise, this might introduce 'collider' bias.

  *"It is not advisable to adjust the main analyses for covariates measured after randomisation because they may be affected by the treatments."* [10]

- Pre-specification of which baseline covariates should be adjusted for, and how, is strongly recommended (i.e., using or not using interaction terms, categorization, splines, etc...).

  *"Sponsors should prospectively specify the detailed procedures for executing covariate-adjusted analysis before any unblinding of comparative data. FDA review will emphasize the prespecified primary analysis rather than post-hoc analyses using different models or covariates."*[11]

[10] EMA scientific guidelines: *ICH Topic E 9 Statistical Principles for Clinical Trials*, 1998, https://www.ema.europa.eu/en/documents/scientific-guideline/ich-e-9-statistical-principles-clinical-trials-step-5_en.pdf

[11] FDA scientific guidelines: "Adjusting for Covariates in Randomized Clinical Trials for Drugs and Biological Products Guidance for Industry", 2023, accessed 29/10/2025, https://www.fda.gov/media/148910/download

# Further remarks (2/2)

- Changes to pre-specified analysis choices should be avoided, as they can lead to suspicions of subjective decisions that could increase type-I error.

- But, minor or well argumented changes in how baseline covariates will be used in the analysis (or other 'non major' change) is usually considered acceptable before unblinding the data, for double-blind RCTs.

*"A common practice prior to data analysis is to have a blinded data review (BDR). The BDR typically takes place after most of the data are entered into the database, but not too late to make changes, that is, there is still time to allow last minute revision of the statistical methods or data presentations prior to unblinding. During the BDR, the programmer runs the entire set of tables on the blinded data (e.g., by applying a dummy treatment code). The tables are reviewed, and issues such as data distributions, outliers, and so on are evaluated. The SAP may be revised based on these evaluations noting that this is still prior to unblinding."*[12]

*"The statistical analysis plan (see Glossary) may be written as a separate document to be completed after finalising the protocol. In this document, a more technical and detailed elaboration of the principal features stated in the protocol may be included (see section 7.1). The plan may include detailed procedures for executing the statistical analysis of the primary and secondary variables and other data. The plan should be reviewed and possibly updated as a result of the blind review of the data (see 7.1 for definition) and should be finalised before breaking the blind. Formal records should be kept of when the statistical analysis plan was finalised as well as when the blind was subsequently broken."*[13]

[12] Page 203 in *Fundamental Concepts for New Clinical Trialists*, by Scott Evans & Naitee Ting (2015)

[13] EMA scientific guidelines: *ICH Topic E 9 Statistical Principles for Clinical Trials*, 1998, `https://www.ema.europa.eu/en/documents/scientific-guideline/ich-e-9-statistical-principles-clinical-trials-step-5_en.pdf`

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Covariate adjustment

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Missing data, with a focus on baseline follow-up studies

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# Missing values in baseline follow-up study (Case in neurodegenerative disease)

Data: scores of $n$=166 patients randomized 1:1, some missing values at follow-up visits (missed visit). The higher the score the better. Shown are changes from baseline.

| | id | trt | baseline | week6 | week12 |
|---|---|---|---|---|---|
| 1 | 1 | SoC | 1.4149135 | 0.5570839 | 0.4874180 |
| 2 | 2 | SoC | 0.5392197 | 0.6747093 | 0.2686820 |
| 3 | 3 | Exp | 0.6554562 | -0.7778319 | 0.6444571 |
| 4 | 4 | Exp | 1.7226614 | 2.2641281 | 0.7723273 |
| 5 | 5 | SoC | -2.8416278 | 0.9717710 | 2.1931570 |
| 6 | 6 | SoC | 2.7684744 | -2.5536338 | NA |



Research question: Does the experimental treatment (Exp) improve the score of the patients at week 12, as compared to standard of care (SoC)?

**Source:** simulated data similar to that of a study sponsored by H. Lundbeck A/S, see Baayen et al., Stat.Med., 2025, DOI: 10.1002/sim.70130.

# Missing data are challenging!

**PRINCIPLES**

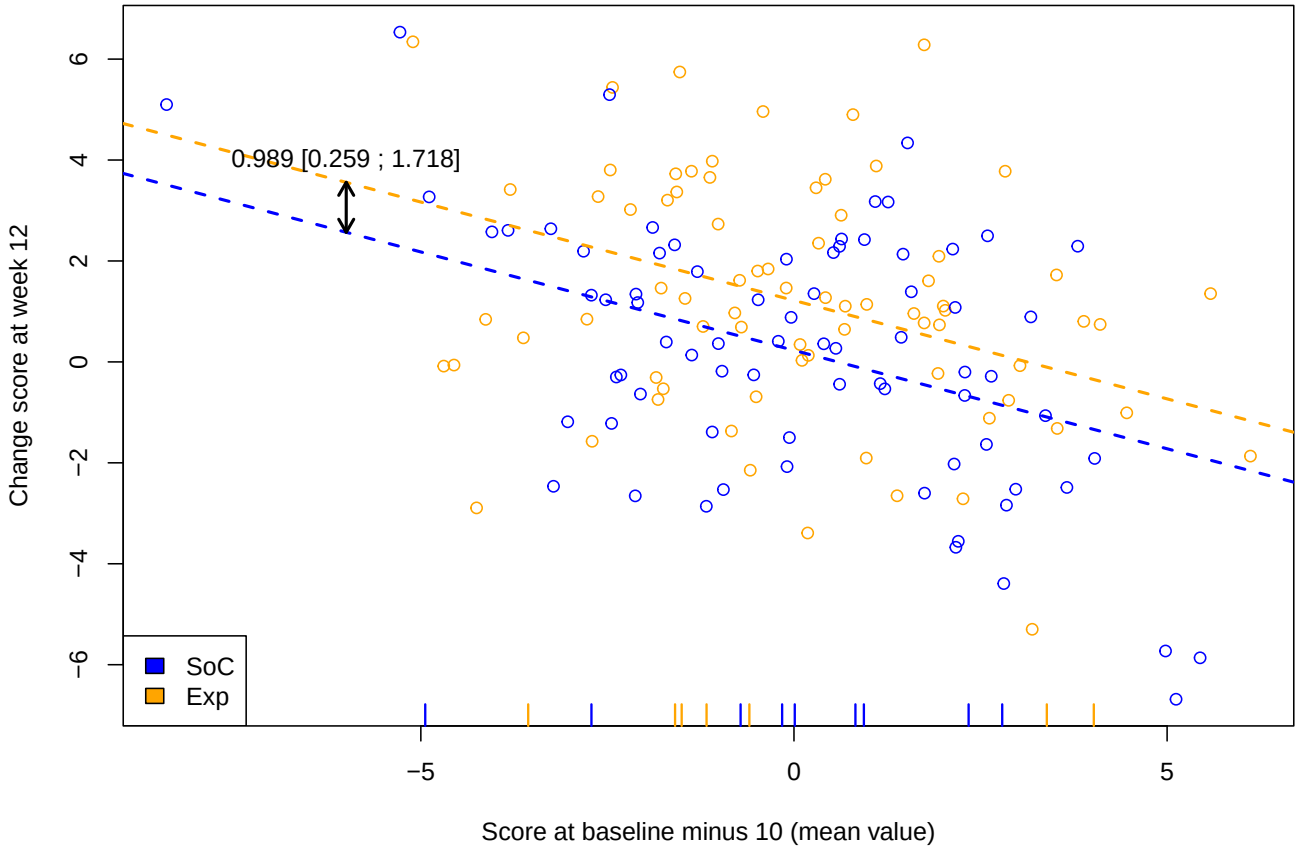
There is no universal method for handling incomplete data in a clinical trial. Each trial has its own set of design and measurement characteristics. There is, however, a set of six principles that can be applied in a wide variety of settings.

- *"First, it needs to be determined whether missingness of a particular value hides a true underlying value that is meaningful for analysis."* E.g., Quality of Life, pain or functioning scores or CD4 counts do not exist after death!
- *"Third, reasons for missing data must be documented as much as possible."*
- *"Fourth, the trial designers should decide on a primary set of assumptions about the missing data mechanism."*
- *"Fifth, the trial sponsors should conduct a statistically valid analysis under the primary missing data assumptions."*

Note: the second principle is about well-defined estimands, the sixth about sensitivity analysis.

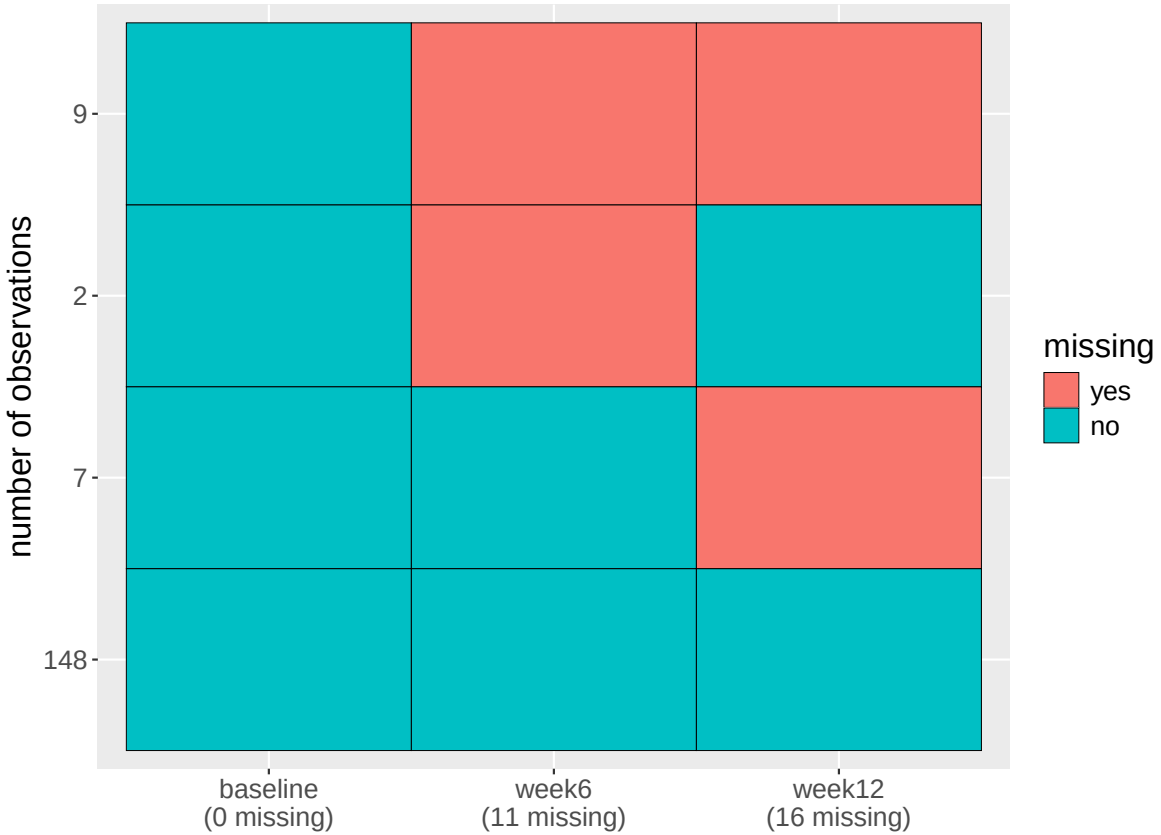
**Source:** National Research Council. 2010. *The Prevention and Treatment of Missing Data in Clinical Trials*. Washington, DC: The National Academies Press. https://doi.org/10.17226/12955 (pages 48-49)






- Without missing data, the "standard" approach is to fit a usual ANCOVA model. It would provide "model robust" conclusions thanks to randomization.
- Results shown are those obtained from the "complete case analysis" (i.e., excluding patients with a missing change score at 12 weeks).
- Baseline scores of the 16 patients with missing values for the change score at week 12 are shown by "ticks" on the x-axis.

# Missing data pattern




It is usually useful to transparently report the missingness pattern and explain what could be the most likely causes of each case. Recommendation: always produce this plot, at least to check that there is nothing unexpected or implausible.

# Missing data: issues and a solution (often useful, but not always)

Challenges with missing data include:

- if the missing data are not "Missing Completely At Random" (MCAR), the complete case analysis is usually biased. Informally, we say that the missing data are MCAR if missingness is unrelated to outcome and covariates.
- even with MCAR, using some information available about the excluded patients usually increases the power (e.g., change score at week 6, whenever available but that at week 12 is missing). Idea: using some kind of information is better than none!

Solution using a Mixed-effect Model for Repeated Measures (MMRM):

- prevents bias if the data are "Missing At Random" (MAR), meaning that the missingness may depend on covariates and previous measures of the outcome (e.g., change score at week 6), but is otherwise completely random.
- more powerful in case of MCAR.



# The MMRM model

Model for the outcome "change score" at the $j$-th visit ($j$=1 for visit at week 6; $j$=2 for visit at week 12) of the $i$-th patient:

$$Y_{ij} = \mu + \mathbf{x}_i\beta_1 + \mathbf{z}_j\beta_2 + \mathbf{u}_i\beta_3 + \mathbf{x}_i \cdot \mathbf{z}_j\beta_4 + \mathbf{u}_i \cdot \mathbf{z}_j\beta_5 + \varepsilon_{ij}$$

with

$$\varepsilon_{ij} \sim \mathcal{N}\left( \begin{bmatrix} 0 \\ 0 \end{bmatrix} , \begin{bmatrix} \sigma_1^2 & \rho\sigma_2\sigma_1 \\ \rho\sigma_2\sigma_1 & \sigma_2^2 \end{bmatrix} \right)$$

- $\mathbf{x}_i$: baseline score of the $i$-th patient minus 10 (the "average" baseline score).
- $\mathbf{z}_j$: indicates the week, equal 1 if j=1 (week 6), 0 if j=2 (week 12)[14]
- $\mathbf{u}_i$: indicates the arm, equal 1 if $i$-th patient randomized to "Experimental" arm, 0 if randomized to "Standard of Care".

- Observations from different patients are assumed to be independent.
- The name "MMRM" is because this model is an extension of the random effect/mixed model which assumes $\sigma_1 = \sigma_2 = \sigma_T$.





[14] **Important trick:** set the reference to the visit at which we observe the primary outcome.

## R code:

```
library(LMMstar)
lmmfit <- lmm(score~baseline*visit + trt*visit, repetition = ~visit|id,
              structure = "UN", data = long)
summary(lmmfit)
```

## Output (partial):

```
Residual variance-covariance: unstructured

  - correlation structure: ~0 + visit
        2     1
  2 1.000 0.627
  1 0.627 1.000

  - variance structure: ~visit
          standard.deviation ratio
  sigma.2               2.25 1.000
  sigma.1               2.04 0.908

Fixed effects: score ~ baseline * visit + trt * visit

                  estimate    se    df  lower  upper p.value
  (Intercept)        0.226 0.258 152.5 -0.283  0.735 0.38236
  baseline          -0.395 0.072 154.2 -0.537 -0.252 < 1e-04 ***
  visit1            -0.083 0.216   147  -0.51  0.343 0.69972
  trtExp             0.984 0.364   153  0.265  1.703 0.00766  **
  baseline:visit1    0.157 0.061 148.9  0.036  0.277 0.01099   *
  visit1:trtExp     -0.497 0.305 147.3 -1.099  0.105 0.10513
```

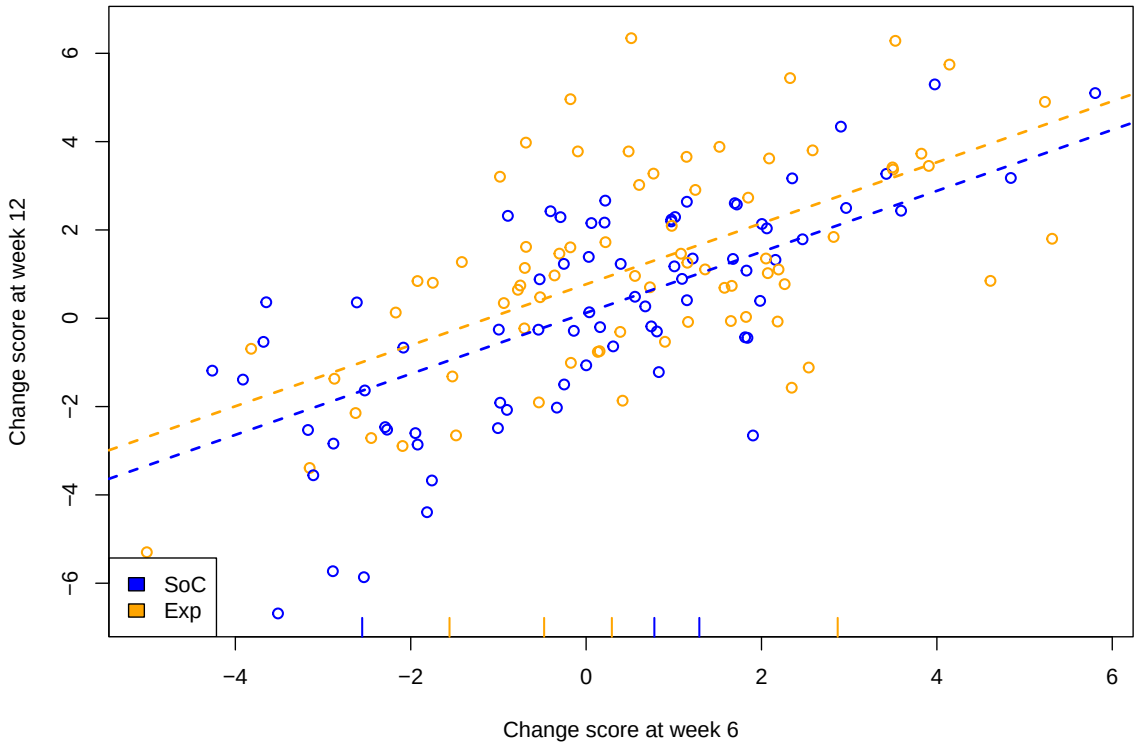


# Parameters interpretation (reference group: trt=SoC, visit=2 (12 weeks) and baseline score - 10=0)

- $\mu$: estimated as 0.226, is the mean change score at 12 weeks for a patient randomized to "SoC", with baseline score=10 (because we have removed 10 to all baseline scores, before fitting the model).
- $\beta_1$: estimated as -0.395, is the mean change in outcome (change score at 12 weeks) when comparing two patients of the same arm, one with a baseline score one unit larger than the other.
- $\beta_2$: estimated as -0.083, is the mean difference between the change score at week 6 and at week 12, for a patient of baseline score=10 and randomized to "SoC".
- $\beta_3$: estimated as 0.984, is the mean difference in outcome (change score at 12 weeks) for a patient randomized to "Exp" as compared to a patient randomized to "SoC", when both patient have the same baseline score (or "in average", due to randomization).
- $\beta_4$: estimated as 0.157, is a difference of differences in mean. In short, the difference $\beta_1$ becomes $\beta_1 + \beta_4$ when comparing change scores at week 6 instead of change scores at week 12. It is just to let the data speak freely and model a possibly different association between the baseline score and the change scores at 6 and 12 weeks.
- $\beta_5$: estimated as -0.497, is a difference of differences in mean. In short, the difference $\beta_2$ becomes $\beta_2 + \beta_5$ for a patient randomized to "Exp". It is just to let the data speak freely and model possibly different treatment effects on the change scores at 6 and 12 weeks.
- $\sigma_1$: estimated as 2.04, is the standard deviation of the "unexplained" variability of the change score at 6 weeks (i.e., standard deviation of error term $\varepsilon_{i1}$; unexplained because neither explained by the treatment nor by the baseline score; prediction interval is "estimated mean" $\pm 1.96\sigma_1$; see plot on next slides).
- $\sigma_2$: estimated as 2.25, same interpretation as for $\sigma_1$, but for the change score at 12 weeks.
- $\rho$: estimated as 0.627, is the correlation between the change score of the same patient at 6 and 12 weeks (see plot on next slide).

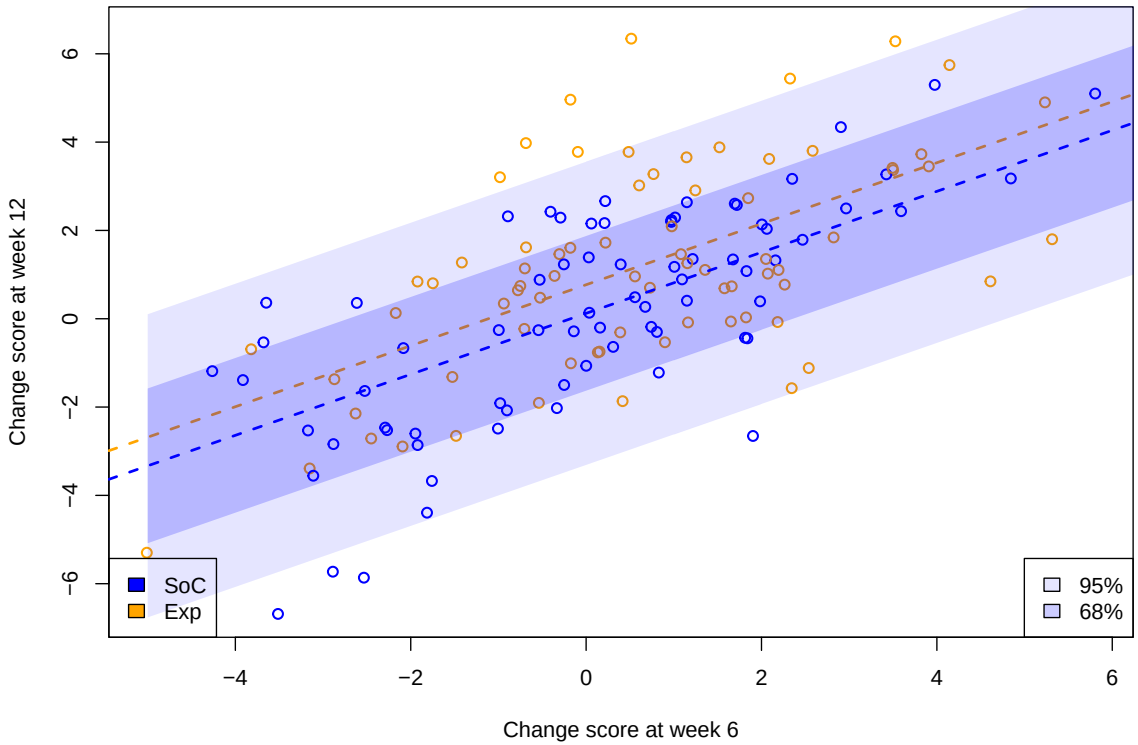


# How does the MMRM handle missing data? (1/2)




- The lines show the estimated average change score at 12 weeks for a patient of baseline score=10 (the average value), for both arms (lines would be shifted up or down for other baseline scores).
- The slope (assumed to be the same in the two groups) is the estimated value of $\rho \cdot \sigma_2/\sigma_1$
  (if $\sigma_2 \approx \sigma_1$, then slope $\approx \rho$, i.e., the correlation between the change score of the same patient at 6 and 12 weeks)
- The change score at week 6 of the 7 patients with missing values for the change score at week 12 (but no missing value at week 6) are shown by "ticks" on the x-axis.

# How does the MMRM handle missing data? (2/2)




- Shown are 68% and 95% prediction intervals for the change score at week 12, given an observed value of the change score at week 6 and baseline score=10, in the standard of care arm (intervals would be shifted up or down for other baseline values and/or treatment arm).
- Implicitly, the MMRM "guesses" the likely values of the missing scores at 12 weeks, given the available information at baseline and week 6. Because the "guesses" use this information, the results will be robust to missingness mechanisms that depend on baseline score and change score at week 6, which is more realistic than assuming that it depends only on what is observed at baseline (which is what the complete case analysis using ANCOVA assumes). This will also typically lead to power gains, when data are MCAR. This is intuitive. E.g., if correlation $\rho \approx 1$, then knowing the change score at week 6 is almost as good as knowing it at week 12, so not much loss of information hence not much loss of power.
- Paralell lines assumption can be relaxed by using more advanced methods, if necessary, see e.g., Wang & Du Int J Biostat (2023).

# MMRM: further remarks

- Including additional (prognostic) baselines covariate can increase precision/power (e.g, sex, age, disease severity, in addition to (or instead of) the baseline score). This is common practice in the analysis of RCT data (see e.g., practical with knee surgery data).

- Although including interactions between baseline covariate and visits/follow-up times is not always necessary, it is usually recommended.[15] This provides:
  - more power/precision
  - lead to similar results to that of a usual ANCOVA analysis when there is no missing data.

  Hence, although the modeling is more complicated (and so is the software output), the interpretation is "simpler", because it provides a method similar to the usual ANCOVA except for the (better) handling of missing data.





[15] See e.g., Schuler Ther. Innov. Regul. Sci. (2022) or Wang & Du Int J Biostat (2023) and references therein.

# Why are MMRM commonly used?

- Most modeling assumptions are not so important for the analysis of (sufficiently large) randomized data.[16]
  - E.g., the assumption that the error terms are normally distributed isn't important unless sample sizes are small. Linear mixed models are highly robust due to the central limit theorem.
- The MAR assumption is often more realistic than the MCAR assumption. The MAR assumption is sufficient[17] to avoid bias and it is difficult to use another modeling approach without assuming MAR (at least a "simple" alternative; complementary sensitivity analyses can be useful).
- User-friendly software exist (and most are also reliable/validated).
- Many reliable and consensual guidelines and textbooks recommend MMRM as a good "default choice" in many common contexts.[18]

[16] See e.g., Wang et al. (2021). J. Am. Stat. Assoc., 118(542), 1152–1163 and Wang & Du Int J Biostat (2023).

[17] Precisely, it is sufficient under "correct" model specification.

[18] See e.g., Mallinckrod et (2008), *"Recommendations for the Primary Analysis of Continuous Endpoints in Longitudinal Clinical Trials,"* Drug Information Journal, 42, 303–319.

# Exercise 3.2

1. Practice with R! Run the code of the script "`neuroRDemo.R`" and recognize the main results of the previous slides.

2. Do the complete case analysis, using an ANCOVA to adjust for baseline. How different are the results? Hint: to create the complete case data, you can use: `dCCA <- d[!is.na(d$week12),]`.
3. Fit the MMRM on the same complete data. What do you observe? Is it reassuring?



# Exercise 3.3

Practice with the "Exercise.3.3" from the webpage.
It is about a knee surgery trial.



vs



**Note:** the (simulated) data of this exercise and their analysis closely ressemble that of a recent investigator-initiated Danish RCT described in:

- Protocol paper: Mortensen et al (2019), BMC Musculoskeletal Disorders, `https://doi.org/10.1186/s12891-019-2508-1`.
- Results at 2 years: Mortensen et al, Journal of Bone and Joint Surgery, 2026 `http://dx.doi.org/10.2106/JBJS.25.00612`.

But we recommend that you do not look at the papers before finishing the exercise (especially not at the seond).

# Missing baseline covariates

- *"Missing covariate values are not inevitable in randomised trials but do sometimes occur.* [...] *Unlike with missing outcome data, discarding individuals with observed data on treatment and outcome does not follow the intention-to-treat principle and does throw away information."* [19]

- *"A simple approach to missing baseline values is to replace them by the mean of the observed baseline values.* [...] *missing baselines should not be imputed within arms"* [20]

- *"The method of dealing with missing baseline covariates should be specified before trial data are analysed* [...] *pre-specification of a sensible method is more important than finnding the perfect method"*[20]

[19] Morris et al. "Planning a method for covariate adjustment in individually randomised trials: a practical guide." Trials 23.1 (2022): 328.

[20] White & Thompson. Adjusting for partially missing baseline measurements in randomised trials. Stat Med. 2005;24(7):993–1007